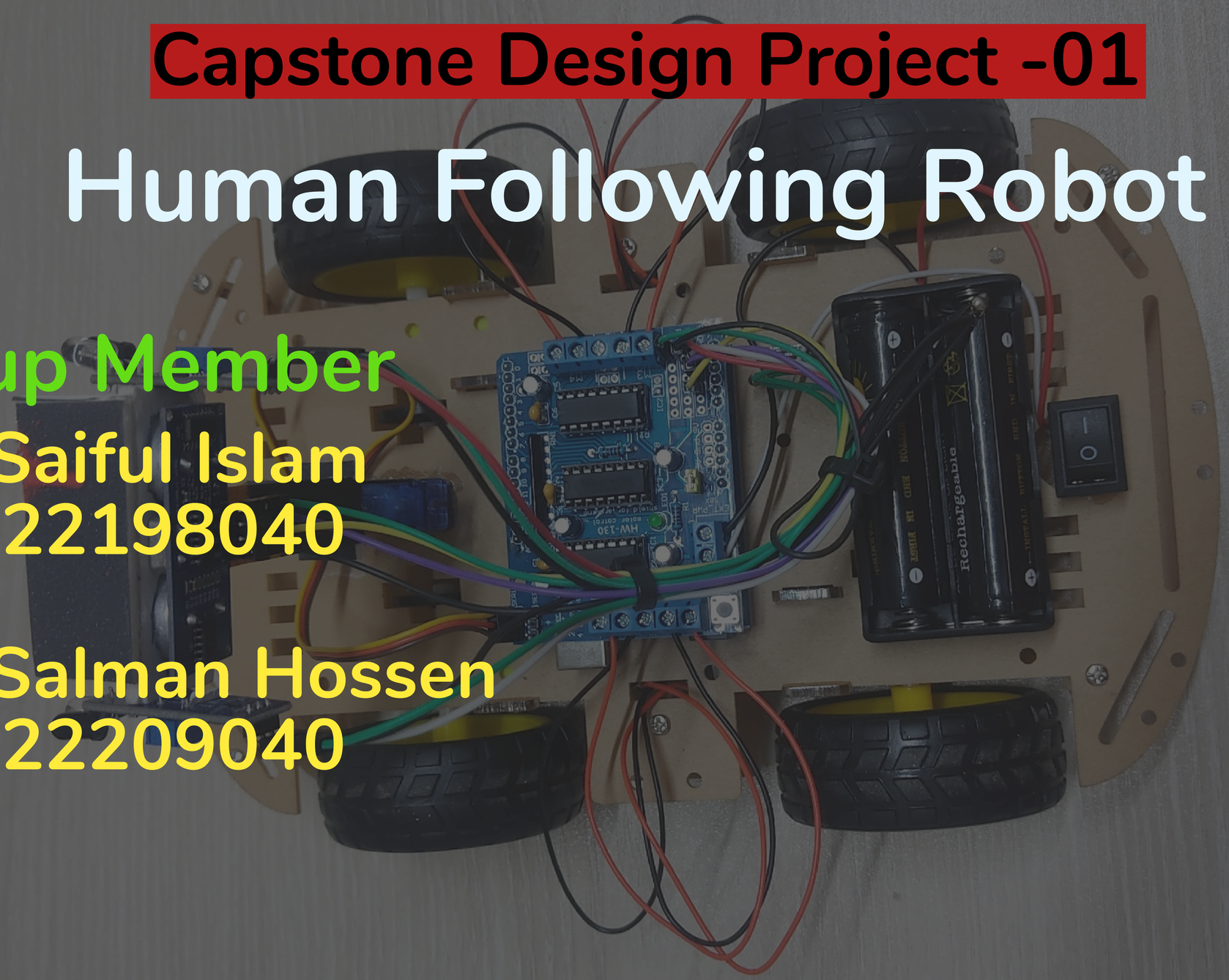
# Capstone Design Project -01

# Human Following Robot





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# Introduction :

The Human Following Robot is an Autonomous Mobile Platform designed to detect, track, and follow a person without direct manual control.

Its core function is to maintain a safe, fixed distance from the human target while autonomously navigating. The robot's intelligence relies on sensor fusion: an Ultrasonic Sensor (HC-SR04) measures the distance to control speed, and IR Sensors provide directional data for smooth differential steering (turning left/right).

Built around the Arduino UNO and a motor shield, this project demonstrates real-time object tracking and collision avoidance, making it applicable for roles in personal assistance or educational robotics.

## Components :

*Ultrasonic sensor
*Motor driver shield
*Arduino Uno
*Servo motor
*Infrared sensor
*Dc gear motor
*Car kit
*Lithium-ion Battery (3.7v)
*Battery holder
*Switch
*Wires

# Components in Brief :

Arduino Uno :- The Arduino Uno is an open-source, flexible, and easy-to-use microcontroller board based on the ATmega328P.

It features digital and analog input/output pins, making it versatile for a variety of electronic projects, from simple LED control to robotics and IoT devices.

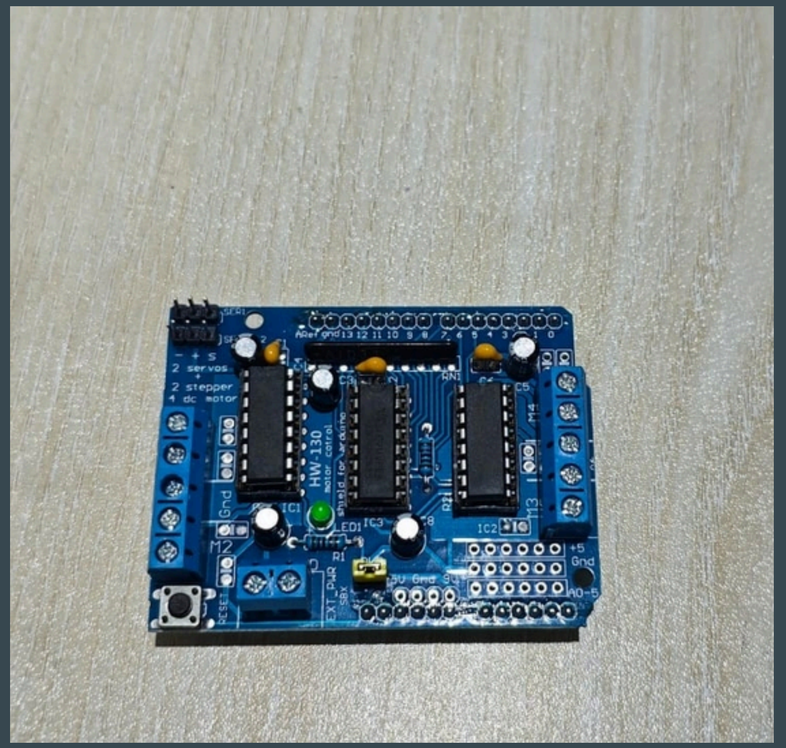


# Motor Driver Shield :

A motor driver shield is an expansion board that simplifies the process of controlling motors with a development board

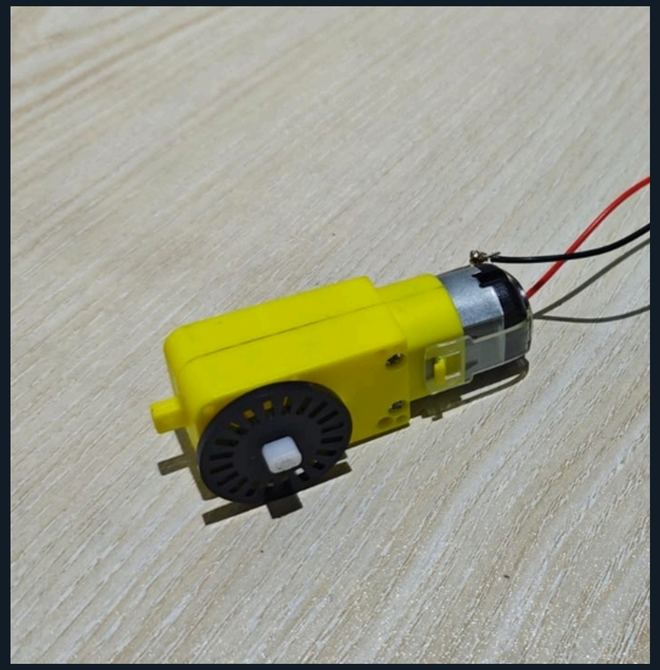
like the Arduino Uno by providing the necessary power amplification and control circuitry in a convenient, stackable format.

The most popular option is the L293D Motor Driver Shield.



# DC Gear Motor :

The DC TT gear motor is an affordable, durable geared DC motor widely used in hobbyist robotics and smart car projects due to its simplicity within the 3V-6V operating range.



It features a built-in gearbox that increases torque while reducing speed, making it suitable for driving wheels and small mechanisms.

# Servo Motor :

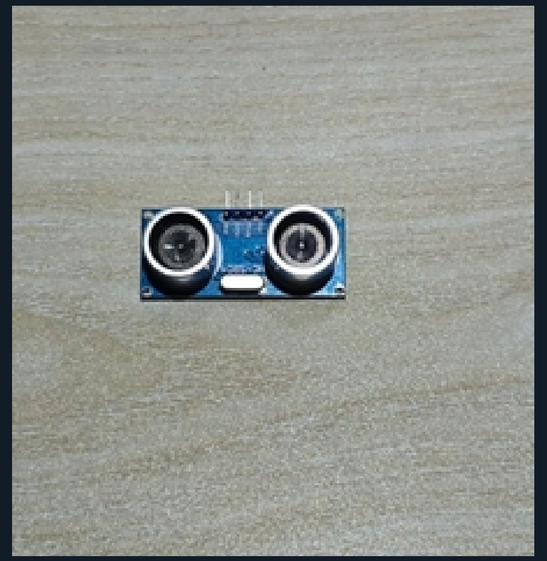
★ A servo motor is a self-contained electrical device that uses a DC motor, gear train, control circuit, and built-in position feedback mechanism (potentiometer) to rotate a shaft to specific angular positions with high precision.



★ They are commonly used in robotics, remote-controlled models, and automation systems where precise control over angular position is required.

# Ultrasonic Sensor :

★ An ultrasonic sensor is a device that measures the distance to an object using sound waves.



★ It works on a principle similar to sonar, emitting a high-frequency sound wave (typically 40 kHz, beyond human hearing) from a transmitter and then calculating the time it takes for the echo to return to the receiver.

# Infrared Sensor :

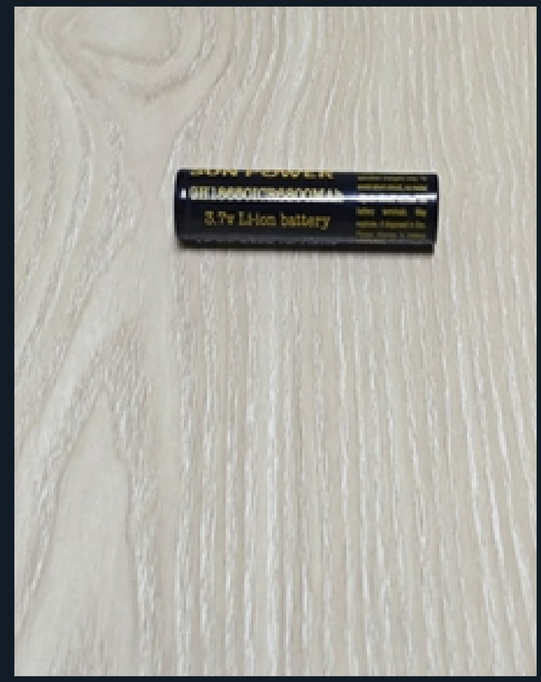
★ An infrared (IR) sensor is a device that detects infrared radiation emitted by an object or the environment.

★ These sensors are detecting ambient IR light or by actively emitting IR light and measuring the reflection or interruption.



# Lithium-ion Battery :

A lithium-ion (Li-ion) 18650 battery is a widely used, rechargeable cylindrical cell known for its high energy density and nominal voltage of 3.7V.




The name "18650" refers to its physical dimensions: 18mm in diameter and 65mm in length.

# Circuit Connection :

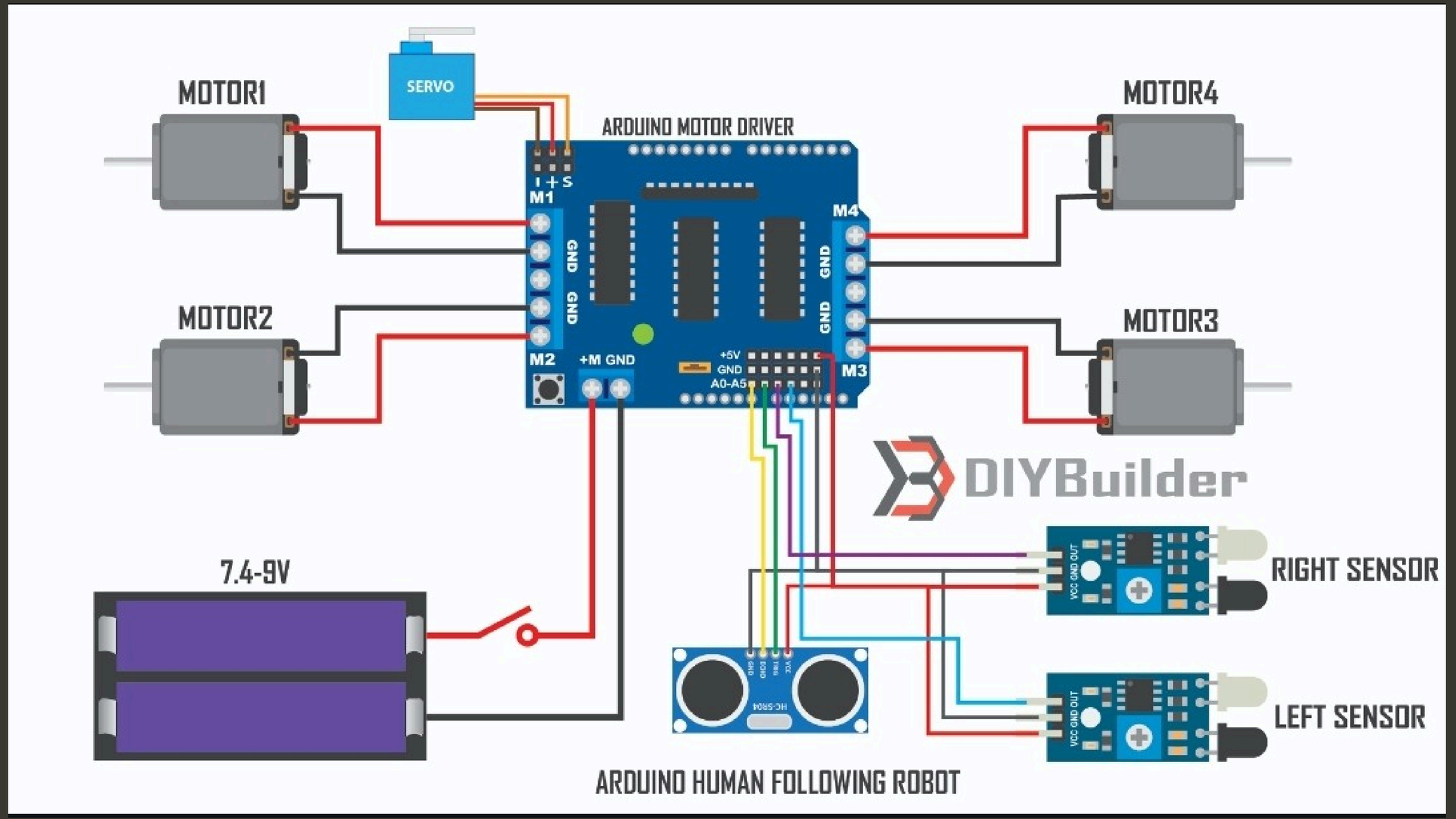
1.The circuit uses an Arduino with a motor shield. A 7.4-9V battery powers the shield.

2.Four DC motors are connected to M1-M4 terminals.

3.An ultrasonic sensor's Trigger and Echo pins connect to Arduino pins A1 and A0 (Digital 9 and 8).

4.The Right and Left IR sensor outputs connect to Arduino pins A3 and A2 (Digital 11 and 10). A servo connects to the dedicated 'SERVO' header (Pin 10)

# Circuit Diagram :



# Parts of Code :

If the robot detects an obstacle closer than 20cm, the code enters a specific block to stop, back up, and then stop again.

```
void loop() {
  int distance = getDistance(); // Get

  // ... (IR sensor code) ...

  Serial.print("Distance: ");
  Serial.print(distance); // Print dis

  if (distance < 20) { // Check if an
    stopMotors();
    // ... (backward movement logic to
  }
  // ... (rest of the movement logic)
}
```

# Application :

1.to assist humans by carrying items and reducing physical strain.

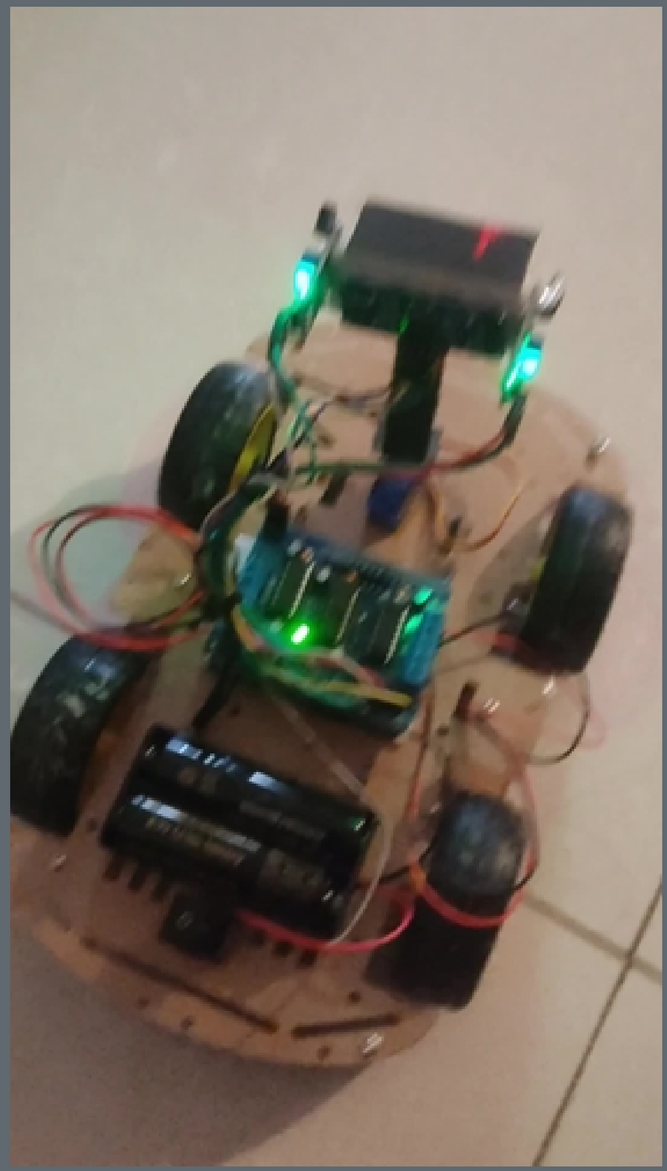
2.Luggage Transport at Airports/Stations

3.Elderly and Disabled Assistance

4.Shopping Centers and Retail



# Working Output :



# Thanks Everyone